National University of Computer and Emerging Sciences, Lahore

Signals and Systems (EE2008) — Assignment 3 — CLO #4

BS Electrical Engineering Section: BEE-4D Spring 2026 Total Marks: 48

Student Name: ______________________ Roll No: ______________ Section: ________

Question 1 — Energy of the signal (Fig. 1)

The discrete signal is the triangular pulse

$$x[n] = \{\,1,\ 2,\ 3,\ 2,\ 1\,\} \quad \text{at } n = 1,2,3,4,5, \qquad x[n] = 0 \text{ elsewhere.}$$

For a finite-energy (aperiodic) signal the energy is

$$E = \sum_{n=-\infty}^{\infty} \left|x[n]\right|^2 = 1^2 + 2^2 + 3^2 + 2^2 + 1^2 = 1 + 4 + 9 + 4 + 1.$$

$$\boxed{E = 19 \text{ (energy units / joules)}}$$

Since E is finite and non-zero, $x[n]$ is an **energy signal** (its power is 0).

Question 2 — Power of the signal (Fig. 2)

The signal is **periodic** with fundamental period $N = 6$ (the peaks of height 3 repeat every 6 samples). The impulse magnitudes given are $0, 1, 2, 3$, so one period is

$$x[n] = \{\,3,\ 2,\ 1,\ 0,\ 1,\ 2\,\} \quad \text{(equivalently } 0,1,2,3,2,1), \qquad N = 6.$$

For a periodic signal the average power is

$$P = \frac{1}{N} \sum_{n=\langle N \rangle} \left|x[n]\right|^2 = \frac{1}{6}\left(3^2 + 2^2 + 1^2 + 0^2 + 1^2 + 2^2\right) = \frac{9+4+1+0+1+4}{6} = \frac{19}{6}.$$

$$\boxed{P = \frac{19}{6} \approx 3.17 \text{ (power units / watts)}}$$

Because P is finite and non-zero (and $E \to \infty$), $x[n]$ is a **power signal**.

Question 3 — Transformations of $x[n]$ (Fig. 3)

Here $x[n] = \{1, 2, 3, 2, 1\}$ on $n = 1, \ldots, 5$ (peak 3 at $n = 3$). For each operation we map indices and tabulate the nonzero samples; the sketches follow.

(a) $x[-n]$ — time reversal about $n = 0$:
nonzero where $-n \in \{1, \ldots, 5\}$, i.e. $n = -1, \ldots, -5$: $x[-n] = \{1, 2, 3, 2, 1\}$ at $n = -1, -2, -3, -4, -5$ (peak at $n = -3$).

(b) $x[n + 6]$ — advance (shift left) by 6:
nonzero where $n + 6 \in \{1, \ldots, 5\}$, i.e. $n = -5, \ldots, -1$: peak at $n = -3$. *Note:* since $x[n]$ is symmetric about $n = 3$, this coincides exactly with $x[-n]$.

(c) $x[n - 6]$ — delay (shift right) by 6:
nonzero where $n - 6 \in \{1, \ldots, 5\}$, i.e. $n = 7, \ldots, 11$: $\{1, 2, 3, 2, 1\}$ at $n = 7, 8, 9, 10, 11$ (peak at $n = 9$).

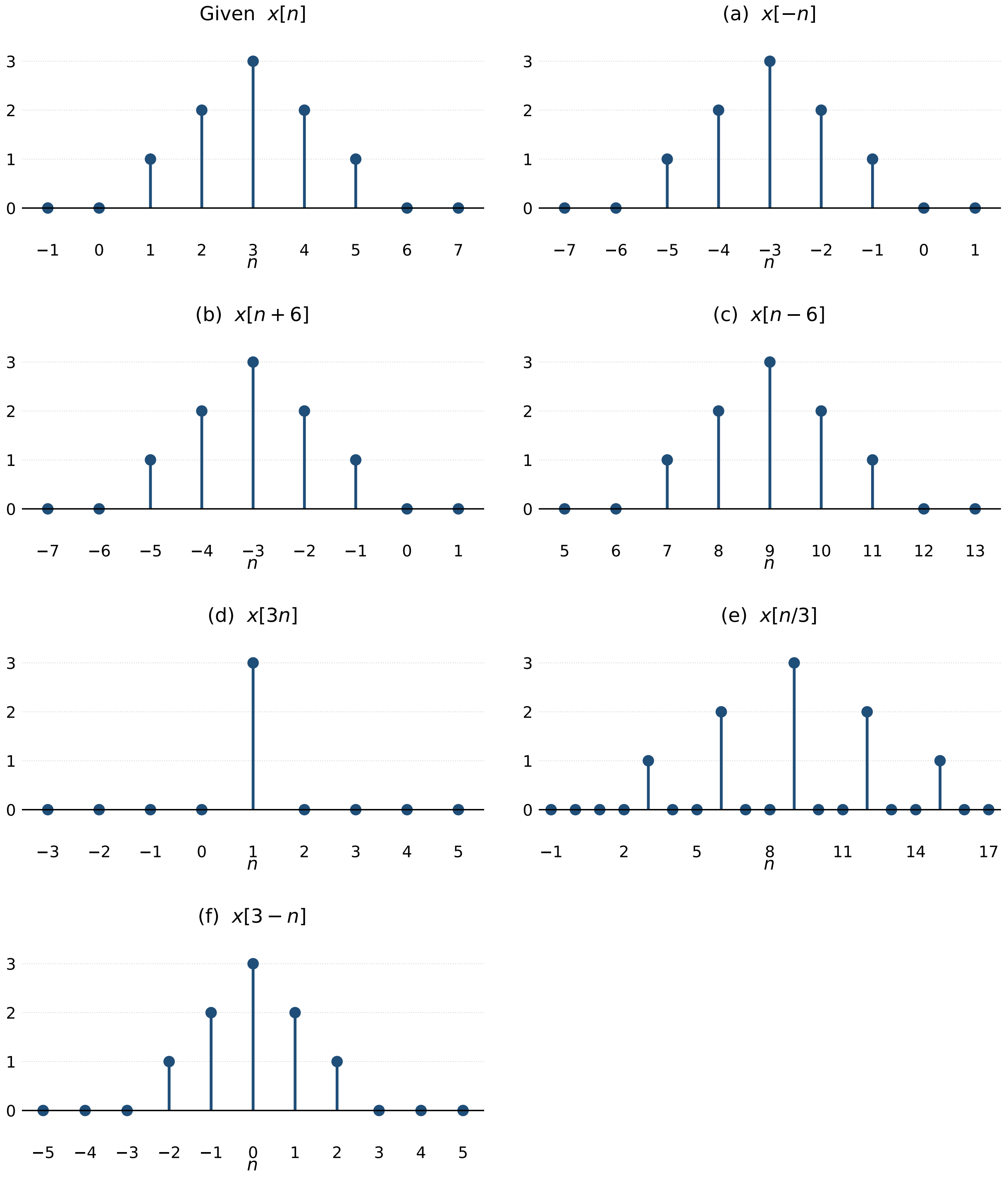
(d) $x[3n]$ — downsampling (compression) by 3:
$3n$ must lie in $\{1, \dots, 5\}$ for an integer n. Only $3n = 3 \Rightarrow n = 1$ qualifies, giving $x[3] = 3$. Hence

$$x[3n] = 3\,\delta[n-1] \quad \text{(a single sample of height 3 at } n = 1\text{)}.$$

Most samples are lost — this is the data loss caused by decimation.

(e) $x[n/3]$ — upsampling (expansion) by 3:
defined only when $n/3$ is an integer; nonzero where $n/3 \in \{1, \dots, 5\}$, i.e. $n = 3, 6, 9, 12, 15$: $\{1, 2, 3, 2, 1\}$ at $n = 3, 6, 9, 12, 15$, with zeros inserted between (peak at $n = 9$).

(f) $x[3-n]$ — reversal *and* shift ($x[-(n-3)]$):
nonzero where $3 - n \in \{1, \dots, 5\}$, i.e. $n = 2, 1, 0, -1, -2$: $\{1, 2, 3, 2, 1\}$ at $n = -2, -1, 0, 1, 2$ (peak at $n = 0$).



Question 4 — BIBO stability and causality

The system (rewriting $y[n+1] = \dfrac{x[n]}{x[n+1]}$ with $n \to n-1$) is

$$y[n] = \frac{x[n-1]}{x[n]}.$$

(i) Causality. The output at time n uses $x[n]$ (present) and $x[n-1]$ (past) only — no future inputs.

$$\boxed{\text{The system is } \textbf{causal}.}$$

(ii) BIBO stability. A system is BIBO stable if *every* bounded input gives a bounded output. The division by $x[n]$ destroys this. Take the bounded input

$$x[n] = \begin{cases} \varepsilon, & n = 0 \\ 1, & n \neq 0 \end{cases} \qquad (|x[n]| \leq 1 \text{ is bounded}).$$

Then $y[0] = \dfrac{x[-1]}{x[0]} = \dfrac{1}{\varepsilon} \to \infty$ as $\varepsilon \to 0$. A bounded input produces an unbounded output, so

$$\boxed{\text{the system is } \textbf{not} \text{ BIBO stable.}}$$

Question 5 — Recursive solution (first three terms)

$$y[n] - 0.6\,y[n-1] - 0.16\,y[n-2] = 0 \;\Rightarrow\; y[n] = 0.6\,y[n-1] + 0.16\,y[n-2],$$

with $y[-1] = -25,\; y[-2] = 0$.

$$\begin{aligned}
y[0] &= 0.6\,y[-1] + 0.16\,y[-2] = 0.6(-25) + 0.16(0) = \mathbf{-15},\\
y[1] &= 0.6\,y[0] + 0.16\,y[-1] = 0.6(-15) + 0.16(-25) = -9 - 4 = \mathbf{-13},\\
y[2] &= 0.6\,y[1] + 0.16\,y[0] = 0.6(-13) + 0.16(-15) = -7.8 - 2.4 = \mathbf{-10.2}.
\end{aligned}$$

$$\boxed{y[0] = -15, \qquad y[1] = -13, \qquad y[2] = -10.2}$$

Question 5 (second part) — Digital differentiator: order and realization

A digital differentiator approximates $\dfrac{d}{dt}x(t)$ by the *first backward difference*:

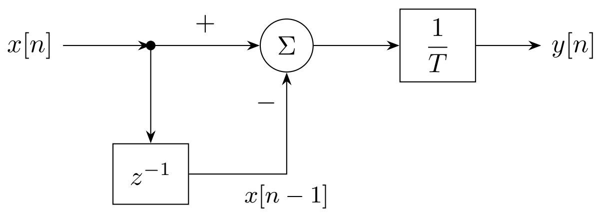
$$y[n] = \frac{x[n] - x[n-1]}{T},$$

where T is the sampling period (with $T = 1$ this is simply $y[n] = x[n] - x[n-1]$).

Order. The highest delay appearing is $x[n-1]$ (one unit delay), and there are no delayed output terms. Therefore the difference equation is **first order** (one delay element), and it is a non-recursive / FIR system.

$$\boxed{\text{Order} = 1}$$

Block-diagram realization.



The input branches: one path enters the adder directly (+), the other is delayed by z^{-1} to form $x[n-1]$ and subtracted (−); the difference is scaled by $1/T$ to give $y[n]$.

Question 6 — z-transforms and ROC (from the definition)

Definition: $X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$. The standard geometric sum $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ for $|r| < 1$ is used throughout.

1) $u[n-m]$ (step starting at $n = m$):

$$X(z) = \sum_{n=m}^{\infty} z^{-n} = z^{-m} \sum_{k=0}^{\infty} z^{-k} = \frac{z^{-m}}{1 - z^{-1}} = z^{-m}\,\frac{z}{z-1}. \qquad \text{ROC: } |z| > 1.$$

2) $\gamma^{n-1}u[n-1]$**:** let $k = n-1$,

$$X(z) = \sum_{n=1}^{\infty} \gamma^{n-1} z^{-n} = z^{-1} \sum_{k=0}^{\infty} \left(\tfrac{\gamma}{z}\right)^k = z^{-1}\frac{z}{z-\gamma} = \frac{1}{z-\gamma}. \qquad \text{ROC: } |z| > |\gamma|.$$

3) $n\,\gamma^n u[n]$**:** using $\gamma^n u[n] \longleftrightarrow \dfrac{z}{z-\gamma}$ and the differentiation property $n\,x[n] \longleftrightarrow -z\dfrac{dX}{dz}$,

$$\frac{d}{dz}\left(\frac{z}{z-\gamma}\right) = \frac{-\gamma}{(z-\gamma)^2} \;\Rightarrow\; X(z) = -z \cdot \frac{-\gamma}{(z-\gamma)^2} = \frac{\gamma z}{(z-\gamma)^2}. \qquad \text{ROC: } |z| > |\gamma|.$$

4) $n\,u[n]$**:** this is part (3) with $\gamma = 1$:

$$X(z) = \frac{z}{(z-1)^2}. \qquad \text{ROC: } |z| > 1.$$

5) $\gamma^n \sin(\pi n)\, u[n]$**:** the general pair is

$$\gamma^n \sin(\beta n) u[n] \;\longleftrightarrow\; \frac{\gamma \sin\beta\, z}{z^2 - 2\gamma\cos\beta\, z + \gamma^2}, \qquad |z| > |\gamma|.$$

Important: for $\beta = \pi$ and integer n, $\sin(\pi n) = 0$ for *all* n. The signal is therefore identically zero, and

$$\boxed{X(z) = 0 \quad \text{(trivial; valid for all } z).}$$

(Substituting $\beta = \pi$ in the general result gives $\sin\pi = 0$ in the numerator, confirming $X(z) = 0$.)

Question 7 — z-transforms using table pairs / properties

Base pairs: $\gamma^n u[n] \longleftrightarrow \dfrac{z}{z-\gamma}$ and the shift property $x[n-n_0] \longleftrightarrow z^{-n_0}X(z)$.

1) $\gamma^{n-2}u[n-2]$**:** delay of $\gamma^n u[n]$ by 2:

$$X(z) = z^{-2} \cdot \frac{z}{z-\gamma} = \frac{z^{-1}}{z-\gamma} = \frac{1}{z(z-\gamma)}. \qquad \text{ROC: } |z| > |\gamma|.$$

2) $2^{n+1}u[n-1] + e^{n-1}u[n]$**:** treat the two terms separately.

Term A: $2^{n+1}u[n-1] = 4 \cdot 2^{n-1}u[n-1]$, and $2^{n-1}u[n-1] \longleftrightarrow z^{-1}\dfrac{z}{z-2} = \dfrac{1}{z-2}$, so

$$\mathcal{Z}\{2^{n+1}u[n-1]\} = \frac{4}{z-2}, \qquad |z| > 2.$$

Term B: $e^{n-1}u[n] = e^{-1}\, e^n u[n]$, and $e^n u[n] \longleftrightarrow \dfrac{z}{z-e}$, so

$$\mathcal{Z}\{e^{n-1}u[n]\} = \frac{1}{e} \cdot \frac{z}{z-e} = \frac{z}{e(z-e)}, \qquad |z| > e.$$

Adding (the overall ROC is the intersection, and $e \approx 2.718 > 2$):

$$\boxed{X(z) = \frac{4}{z-2} + \frac{z}{e(z-e)}, \qquad \text{ROC: } |z| > e \approx 2.718.}$$

Question 8 — Inverse z-transforms

Both share the denominator $z^2 - 5z + 6 = (z-2)(z-3)$. We use $\frac{z}{z-a} \longleftrightarrow a^n u[n]$.

(a) $X(z) = \frac{z(z-4)}{z^2-5z+6}$. Expand $X(z)/z$:

$$\frac{X(z)}{z} = \frac{z-4}{(z-2)(z-3)} = \frac{A}{z-2} + \frac{B}{z-3}.$$

$z - 4 = A(z-3) + B(z-2)$: at $z = 2$, $A = 2$; at $z = 3$, $B = -1$. Hence

$$X(z) = \frac{2z}{z-2} - \frac{z}{z-3} \quad \Rightarrow \quad \boxed{x[n] = \left(2 \cdot 2^n - 3^n\right)u[n] = \left(2^{n+1} - 3^n\right)u[n].}$$

(b) $X(z) = \frac{z-4}{z^2-5z+6}$. This is part (a) divided by z, i.e. a one-sample delay. Directly,

$$X(z) = \frac{z-4}{(z-2)(z-3)} = \frac{2}{z-2} - \frac{1}{z-3},$$

and using $\frac{1}{z-a} = z^{-1}\frac{z}{z-a} \longleftrightarrow a^{n-1}u[n-1]$,

$$\boxed{x[n] = \left(2 \cdot 2^{n-1} - 3^{n-1}\right)u[n-1] = \left(2^n - 3^{n-1}\right)u[n-1].}$$

(Consistently, $x_b[n] = x_a[n-1]$.)

Question 9 — Time-reversal property applied to Pair 6

Pair 6: $\gamma^n u[n] \longleftrightarrow \frac{z}{z-\gamma}$, ROC $|z| > |\gamma|$.

Time-reversal property: if $x[n] \longleftrightarrow X(z)$ with ROC R, then $x[-n] \longleftrightarrow X(z^{-1})$ with ROC $1/R$.

Step 1 — reverse Pair 6. Apply the property to $a^n u[n] \longleftrightarrow \frac{z}{z-a}$ ($|z| > |a|$):

$$a^{-n}u[-n] \longleftrightarrow \frac{z^{-1}}{z^{-1}-a} = \frac{1}{1-az}, \qquad |z| < \tfrac{1}{|a|}.$$

Step 2 — set $a = \frac{1}{\gamma}$ so that $a^{-n} = \gamma^n$:

$$\gamma^n u[-n] \longleftrightarrow \frac{1}{1-\frac{z}{\gamma}} = \frac{\gamma}{\gamma - z} = \frac{-\gamma}{z-\gamma}, \qquad |z| < |\gamma|.$$

Step 3 — convert $u[-n]$ to $u[-(n+1)]$. Since $u[-n] = u[-n-1] + \delta[n]$,

$$\gamma^n u[-(n+1)] = \gamma^n u[-n] - \delta[n] \quad (\gamma^0 = 1).$$

Taking z-transforms ($\delta[n] \longleftrightarrow 1$):

$$\mathcal{Z}\{\gamma^n u[-(n+1)]\} = \frac{-\gamma}{z-\gamma} - 1 = \frac{-\gamma-(z-\gamma)}{z-\gamma} = \frac{-z}{z-\gamma}.$$

$$\boxed{\gamma^n u[-(n+1)] \longleftrightarrow \frac{-z}{z-\gamma}, \qquad \text{ROC: } |z| < |\gamma|.}$$

Check (direct sum): $\sum_{n=-\infty}^{-1} \gamma^n z^{-n} = \sum_{m=1}^{\infty}\left(\frac{z}{\gamma}\right)^m = \frac{z/\gamma}{1-z/\gamma} = \frac{z}{\gamma - z} = \frac{-z}{z-\gamma}$, valid for $|z| < |\gamma|$ — confirming the result.